arc-agi-small-language-model

Taxonomía Deep Learning con Benchmarks ARC-AGI

```
Deep Learning
│
├── 1. Modelos Perceptivos
│   ├── CNNs
│   ├── Vision Transformers (ViT)
│   ├── Multimodal Encoders
│
├── 2. Modelos Generativos
│   ├── GANs
│   ├── VAEs
│   ├── Diffusion Models
│   ├── Flow-based Models
│
├── 3. Modelos Secuenciales
│   ├── RNN / LSTM / GRU
│   ├── Transformers
│   ├── State Space Models (Mamba, S4)
│
├── 4. Foundation Models (Base LLMs)
│   ├── LLMs
│   │   ├── GPT-3 ──────────────────── ARC-AGI-1: ~0%
│   │   ├── GPT-4 ──────────────────── ARC-AGI-1: ~0%
│   │   ├── GPT-4o ─────────────────── ARC-AGI-1: 5%
│   │   ├── GPT-4.1-Nano ───────────── ARC-AGI-1: 0%    | $0.002/task
│   │   ├── Llama 4 Maverick ───────── ARC-AGI-1: 4.4%  | $0.0078/task | ARC-AGI-2: 0%
│   │   ├── Grok 3 ─────────────────── ARC-AGI-1: 5.5%  | $0.0093/task | ARC-AGI-2: 0%
│   │   └── Claude 3.5 Sonnet ──────── ARC-AGI-1: 14% (base, sin técnicas adicionales)
│   ├── Vision Foundation Models
│   └── Multimodal Foundation Models
│
├── 5. Neuro-Symbolic & Reasoning AI
│   ├── Neuro-symbolic models
│   ├── Program Induction Models
│   │   ├── Test-Time Training (MIT/Cornell) ── ARC-AGI-1: 47.5%
│   │   └── CompressARC (76K params) ────────── ARC-AGI-1: 20-34% | ARC-AGI-2: 4% (sin pretraining)
│   ├── Neural Theorem Provers
│   └── Differentiable Reasoning Systems
│
├── 6. World Models & Cognitive Models
│   ├── World Models
│   ├── Predictive Processing Models
│   ├── Active Inference Models
│   └── Cognitive Architectures
│
├── 7. **Reasoning-Centric Models (LRMs - Large Reasoning Models)**
│   │
│   ├── TRM (Token Reasoning Models) - Single Chain-of-Thought
│   │   ├── OpenAI o1 (low) ────────── ARC-AGI-1: 20.5% | $0.43/task
│   │   ├── OpenAI o1 (medium) ─────── ARC-AGI-1: 31%   | $0.79/task
│   │   ├── OpenAI o1 (high) ───────── ARC-AGI-1: 35%   | $1.31/task
│   │   ├── DeepSeek R1-Zero ───────── ARC-AGI-1: 14%   | $0.11/task (sin SFT, solo RL)
│   │   ├── DeepSeek R1 ────────────── ARC-AGI-1: 15.8% | $0.06/task | ARC-AGI-2: 1.3%
│   │   ├── DeepSeek R1 (5/28/25) ──── ARC-AGI-1: 21.2% | $0.046/task (actualizado)
│   │   ├── OpenAI o3 (low) ────────── ARC-AGI-1: 41%   | $1.22/task | ARC-AGI-2: 1.9%
│   │   ├── OpenAI o3 (medium) ─────── ARC-AGI-1: 53%   | $2.52/task | ARC-AGI-2: 2.9%
│   │   ├── OpenAI o3 (high) ───────── ARC-AGI-1: 60.8% | $0.50/task | ARC-AGI-2: 6.5%
│   │   ├── OpenAI o3-pro (high) ───── ARC-AGI-1: 59.3% | $4.16/task | ARC-AGI-2: 4.9%
│   │   ├── OpenAI o4-mini (low) ───── ARC-AGI-1: 21%   | $0.05/task | ARC-AGI-2: 1.6%
│   │   ├── OpenAI o4-mini (medium) ── ARC-AGI-1: 42%   | $0.23/task | ARC-AGI-2: 2.3%
│   │   ├── OpenAI o4-mini (high) ──── ARC-AGI-1: 58.7% | $0.406/task | ARC-AGI-2: 6.1%
│   │   ├── Grok 3 Mini (low) ──────── ARC-AGI-1: 16.5% | $0.0099/task | ARC-AGI-2: 0.4%
│   │   ├── Claude Sonnet 4 (16k) ──── ARC-AGI-1: 40%   | $0.366/task | ARC-AGI-2: 5.9%
│   │   ├── Claude Opus 4 (16k) ────── ARC-AGI-1: 35.7% | $1.25/task  | ARC-AGI-2: 8.6%
│   │   ├── Claude Opus 4.5 (64k) ──── ARC-AGI-2: 37.6% | $2.20/task  | SOTA comercial verificado
│   │   ├── Gemini 2.5 Flash ───────── ARC-AGI-1: 33.3% | $0.037/task (mejor balance)
│   │   ├── Gemini 2.5 Pro ─────────── ARC-AGI-1: 33%   | $0.569/task | ARC-AGI-2: 3.8%
│   │   └── Gemini 3 Pro ───────────── ARC-AGI-2: 31%   | $0.81/task  (baseline)
│   │
│   ├── Search-Augmented Reasoning Models (Test-Time Compute Scaling)
│   │   ├── OpenAI o3-preview (low) ── ARC-AGI-1: 75.7% | $200/task   | 33.5M tokens
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│   │   ├── OpenAI o3-preview (high) ─ ARC-AGI-1: 87.5% | $4,560/task | 5.7B tokens (172x compute)
│   │   └── Claude 3.5 Sonnet + Evolutionary TTC ── ARC-AGI-1: 53.6% (Jeremy Berman)
│   │
│   ├── **Tiny Recursive Models (TRM) - Redes Pequeñas con Razonamiento Recursivo** 🆕
│   │   ├── TRM (7M params) ────────── ARC-AGI-1: 45% | ARC-AGI-2: 8% | Paper Award 1st 2025
(https://arxiv.org/abs/2510.04871)
│   │   │   └── Solo 2 capas, <0.01% parámetros de LLMs, supera DeepSeek R1, o3-mini, Gemini 2.5 Pro
│   │   └── HRM (27M params) ────────── ARC-AGI-1: ~40% | Hierarchical Reasoning Model (precursor)
│   │
│   └── HRM (Hierarchical Reasoning Models)
│       ├── Hierarchical planners
│       ├── Modular reasoning networks
│       └── Multi-level cognitive models
│
├── 8. Agentic AI Systems
│   ├── Autonomous Agents
│   ├── Tool-using Models
│   │   └── SLM Fine-tuned (OPT-350M) ── ToolBench: 77.55% (supera ChatGPT-CoT 26%)
│   ├── Planning Agents
│   └── Multi-agent Systems
│
├── 9. **Soluciones Híbridas / Especializadas (Kaggle ARC Prize)**
│   │
│   ├── ARC Prize 2024 (ARC-AGI-1)
│   │   ├── the ARChitects (1°) ─────── 53.5% (Program synthesis + search)      (https://da-fr.github.io/arc-
prize-2024/the_architects.pdf) usando x4 NVIDIA T4
│   │   ├── Guillermo Barbadillo (2°)── 40%
│   │   ├── alijs (3°) ─────────────── 40%
│   │   ├── William Wu (4°) ────────── 37%
│   │   └── Ensemble Solutions ─────── ~81% (combinación de múltiples soluciones)
│   │
│   └── ARC Prize 2025 (ARC-AGI-2)  (https://arxiv.org/pdf/2601.10904)
│       ├── NVARC (1°) ─────────────── 24.03% | $0.20/task | SOTA Kaggle
(https://drive.google.com/file/d/1vkEluaaJTzaZiJL69TkZovJUkPSDH5Xc/view)
│       ├── the ARChitects (2°) ────── 16.53% | 2D-aware masked-diffusion LLM
(https://lambdalabsml.github.io/ARC2025_Solution_by_the_ARChitects/)
│       ├── MindsAI (3°) ───────────── 12.64% | Test-time training pipeline     (https://arxiv.org/abs/2506.14276)
│       ├── Lonnie (4°) ────────────── 6.67%
(https://www.kaggle.com/competitions/arc-prize-2025/writeups/arc-prize-2025-competition-writeup-5th-place)
│       └── G. Barbadillo (5°) ─────── 6.53%
(https://ironbar.github.io/arc25/05_Solution_Summary/#vision-search-and-learn)
│
└── 10. **Model Refinement Solutions (Refinement Loops)**
    ├── Poetiq (Gemini 3 Pro Ref) ──── ARC-AGI-2: 54% | $31/task | SOTA refinement
    ├── Poetiq (Claude Opus 4.5 Ref)── ARC-AGI-2: ~54% | $60/task
    ├── SOAR (Self-Improving LLM) ──── ARC-AGI-1: 52% | Paper Award 2nd 2025
    └── Evolutionary Program Synthesis (E. Pang) ── Paper Award Runner-up
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Resumen de Benchmarks ARC-AGI (Actualizado Dic 2025)

ARC-AGI-1 Semi-Private Eval (Ranking por rendimiento)

Modelo/Sistema	Score	Costo/Tarea	Notas
o3-preview (high)	87.5%	$4,560	172x compute, 5.7B tokens
Ensemble Kaggle	~81%	-	Combinación de soluciones
o3-preview (low)	75.7%	$200	Primer puesto leaderboard 2024
o3 (high)	60.8%	$0.50	-
o3-pro (high)	59.3%	$4.16	-
o4-mini (high)	58.7%	$0.406	-
Claude 3.5 + EvoTTC	53.6%	-	Evolutionary Test-Time Compute
the ARChitects (2024)	53.5%	-	Ganador Kaggle ARC Prize 2024
o3 (medium)	53%	$2.52	-
SOAR	52%	-	Self-Improving Evolutionary Program Synthesis
TTT MIT/Cornell	47.5%	-	Test-Time Training
TRM (7M params)	45%	-	🆕 Tiny Recursive Model - Paper Award 1st
o3 (low)	41%	$1.22	-

Modelo/Sistema	Score	Costo/Tarea	Notas
Claude Sonnet 4 (16k)	40%	$0.366	-
o4-mini (medium)	42%	$0.23	-
Claude Opus 4 (16k)	35.7%	$1.25	-
o1 (high)	35%	$1.31	-
Gemini 2.5 Flash	33.3%	$0.037	Mejor balance costo/rendimiento
Gemini 2.5 Pro	33%	$0.569	-
o1 (medium)	31%	$0.79	-
DeepSeek R1 (5/28)	21.2%	$0.046	Actualizado
o4-mini (low)	21%	$0.05	-
o1 (low)	20.5%	$0.43	-
CompressARC (76K)	20-34%	-	Sin pretraining, MDL-based
Grok 3 Mini (low)	16.5%	$0.0099	-
DeepSeek R1	15.8%	$0.06	Open source
DeepSeek R1-Zero	14%	$0.11	Sin SFT, solo RL
Claude 3.5 Sonnet	14%	-	Base, sin técnicas adicionales
Grok 3	5.5%	$0.0093	-
GPT-4o	5%	-	Base LLM
Llama 4 Maverick	4.4%	$0.0078	-
GPT-4	~0%	-	Base LLM
GPT-3	0%	-	Base LLM
Humanos	~85%	~$5	Referencia humana

ARC-AGI-2 (lanzado Marzo 2025)

Modelo/Sistema	Score	Costo/Tarea	Notas
Poetiq (Gemini 3 Pro Ref)	54%	$31	🆕 SOTA refinement solution
Claude Opus 4.5 (64k)	37.6%	$2.20	🆕 SOTA comercial verificado
Gemini 3 Pro	31%	$0.81	Baseline
NVARC (Kaggle 1st)	24.03%	$0.20	🆕 SOTA Kaggle 2025
the ARChitects	16.53%	-	2D-aware masked-diffusion
MindsAI	12.64%	-	Test-time training
Claude Opus 4 (16k)	8.6%	$1.93	-
TRM (7M params)	8%	-	🆕 Tiny Recursive Model
o3 (high)	6.5%	$0.834	-
o4-mini (high)	6.1%	$0.856	-
Claude Sonnet 4 (16k)	5.9%	$0.486	-
o3-pro (high)	4.9%	$7.55	-
CompressARC (76K)	4%	-	Sin pretraining
Gemini 2.5 Pro	3.8%	$0.813	-
o3 (medium)	2.9%	-	-
o4-mini (medium)	2.3%	-	-
o3 (low)	1.9%	-	-
o4-mini (low)	1.6%	-	-
DeepSeek R1	1.3%	$0.08	-
Grok 3 Mini (low)	0.4%	$0.013	-

Modelo/Sistema	Score	Costo/Tarea	Notas
Grok 3	0%	$0.14	-
Llama 4 Maverick	0%	$0.012	-
Humanos	>95%	-	Sin entrenamiento previo

Insights Clave (Actualizado 2025)

1. **Los LLMs base fracasan** en ARC-AGI porque dependen de memorización, no razonamiento adaptativo
2. **Los Reasoning Models (o1, R1, o3)** mejoran mediante Chain-of-Thought, pero tienen límites
3. **El test-time compute scaling** permite mejoras logarítmicas pero con rendimientos decrecientes
4. **Tiny Recursive Models (TRM)** logran 45% en ARC-AGI-1 con solo 7M parámetros (<0.01% de LLMs)
5. **ARC-AGI-2 sigue sin resolver**: incluso los mejores modelos <10%, humanos >95%
6. **Refinement Loops** son el tema central de 2025 - mejoran performance iterativamente
7. **La eficiencia importa**: Gemini 2.5 Flash ofrece el mejor ratio costo/rendimiento para ARC-AGI-1
8. **No hay ganador claro**: cada modelo tiene trade-offs entre accuracy, costo y velocidad
9. **ARC-AGI-3** (2026) introducirá razonamiento interactivo: exploración, planificación, memoria

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Bibliografía Relevante

Papers con resultados ARC-AGI

1. **"Less is More: Recursive Reasoning with Tiny Networks"** (Oct 2025)

    - Autor: Alexia Jolicoeur-Martineau
    - arXiv: 2510.04871
    - Resultado: TRM (7M params) → ARC-AGI-1: 45%, ARC-AGI-2: 8%
    - 🏆 Paper Award 1st Place - ARC Prize 2025

2. **"Small Language Models for Efficient Agentic Tool Calling"** (Dec 2025)

    - Autores: Jhandi, Kazi, Subramanian, Sendas
    - arXiv: 2512.15943
    - Resultado: OPT-350M fine-tuned → ToolBench: 77.55% (vs ChatGPT-CoT: 26%)

3. **"On the Measure of Intelligence"** (2019)

    - Autor: François Chollet
    - arXiv: 1911.01547
    - Introduce ARC-AGI benchmark

4. **"The Surprising Effectiveness of Test-Time Training for Abstract Reasoning"** (2024)

    - MIT/Cornell
    - Resultado: ARC-AGI-1: 47.5%

Enlaces imprescindibles

https://lewish.io/posts/arc-agi-2025-research-review#ttt-ttft

https://lewish.io/posts/how-to-beat-arc-agi-2